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4. Development of Bulbils on the rhizoids.

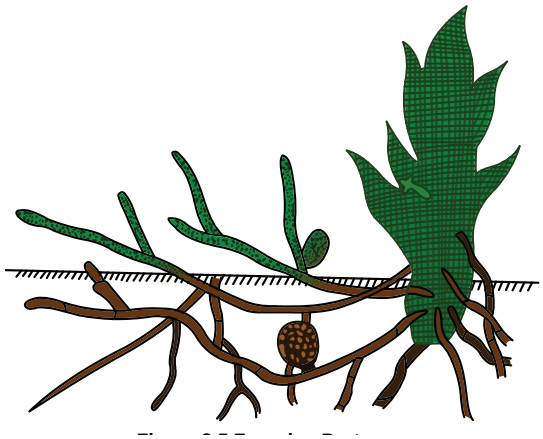
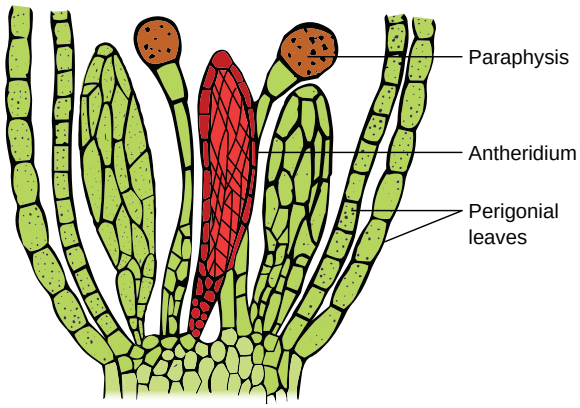


Figure 2.20: Vegetative reproduction (Protonema)

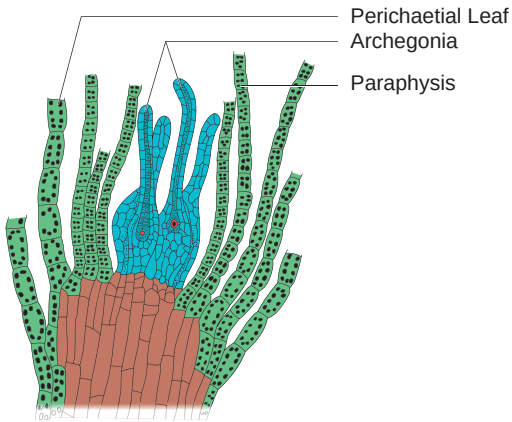
Sexual reproduction

Funaria is monoecious the male and female reproductive sex organs are borne on different branches of the same gametophyte. Male sex organ is antheridium and it is formed in groups on the antheridial branch.They are enclosed by special leaves called **perigonial leaves**. A large number of long multicellular hairs interspersed with antherdia called **paraphysis** are also present. They contain chloroplast and are involved in photosynthesis. They protect antheridial head from by minimizing transpiration, hold water between them through capillary action and secrete mucilage which helps in the liberation of antherozoids. Each antheridium is protected by single layer of jacket. It encloses a large mass of androcytes. The androcytes transform in to biflagellate antherozoids (Figure 2.21).

The female sex organ are the archegonia and are borne in clusters on the archegonial branch. Archegonial branches arise laterally at the base of the male branch. They are surrounded by **perichaetial leaves**. Paraphyses are also present. Each archegonium is flask shaped and is distinguished into a large venter and long neck region. The venter contains venter canal cell and egg. The neck contain neck canal cells (Figure 2.21). Water is essential for fertilization. Rain drops help in the transfer of antherozoids from antheridial head to archegonial head. The antherozoids are attracted to the archegonium through **chemotaxis**. A large number of antherozoids enter the neck of the archegonium but only one fuses with the egg to form a diploid zygote. The diploid zygote represents the first cell of sporophytic generation and divides to form a sporophyte.



a) Antheridial head



b) Archegonial head

Figure 2.21: *Funaria* - Sexual reproduction

Structure of Sporophyte or capsule

The structure of mature sporophyte of *Funaria* is complex. The sporophyte is